LEVEL-II (C.W)

PROBLEMS BASED ON BASIC FORMULA AND SUBSTITUTIONS

1. $\int_{-1}^{1} \frac{d}{dx} Tan^{-1}\left(\frac{1}{x}\right) dx$

1) $\pi/2$ 2) $-\pi/2$
3) π 4) $-\pi$

2. $\int_0^1 \sqrt{\frac{x}{1-x^3}} dx =$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{2}$

3. $\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)} =$

1) $\frac{\pi ab}{(a+b)}$ 2) $\frac{\pi}{2(a+b)}$

3) $\frac{\pi}{2ab(a+b)}$ 4) $\frac{\pi(a+b)}{ab}$

4. **The value of** $\int_1^{\sqrt[7]{2}} \frac{1}{x(2x^7+1)}$ **dx is**

1) log(6/5) 2) 6 log (6/5)

3) (1/7) log (6/5) 4) (1/12) log (6/5)

5. $\int_0^1 \text{Cos}\left(2\text{Cot}^{-1}\sqrt{\frac{1-x}{1+x}}\right) dx =$

1) -1/2 2) 1/2
3) 0 4) 1

6. $\int_{1/3}^{1} \frac{(x-x^3)^{1/3}}{x^4} dx =$

1) 3 2) 0
3) 6 4) 4

7. $\int_0^{\pi/2} \frac{1}{1+4\sin^2 x} dx =$

1) $\frac{\pi}{\sqrt{5}}$ 2) $\frac{\pi}{2\sqrt{5}}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3\sqrt{5}}$

8. $\int_0^{\pi/2} \frac{1}{4+5\cos x} dx =$

1) $\frac{1}{5}\log 2$ 2) $\frac{1}{2}\log 2$ 3) $\frac{1}{3}\log 3$ 4) $\frac{1}{3}\log 2$

9. **If** $\int_0^1 \frac{\tan^{-1}x}{x} dx = k\int_0^{\pi/2} \frac{x}{\sin x} dx$ **then the value of** k **is**

1) 1 2) $\frac{1}{4}$ 3) 4 4) $\frac{1}{2}$

10. **Value of** $\int_1^5 \left(\sqrt{x+2\sqrt{x-1}} + \sqrt{x-2\sqrt{x-1}}\right) dx$ **is**

1) $\frac{8}{3}$ 2) $\frac{16}{3}$ 3) $\frac{32}{3}$ 4) $\frac{34}{3}$

11 **The value of**

$\int_{-2}^{0} \left[x^3+3x^2+3x+3+(x+1)\cos(x+1)\right] dx$ **is**

1) 0 2) 3 3) 4 4) 1

12. $\int_0^{1/2} \frac{x\sin^{-1}x}{\sqrt{1-x^2}} dx =$

1) $\frac{1}{2}+\frac{\pi\sqrt{3}}{12}$ 2) $\frac{1}{2}-\frac{\pi\sqrt{3}}{12}$

3) $\frac{\pi\sqrt{3}}{12}$ 4) $\frac{-\pi\sqrt{3}}{12}$

13. **If** $f(x) = \begin{cases} \sin x & \text{if } 0 \le x < \frac{\pi}{2} \\ 1 & \text{if } \frac{\pi}{2} \le x < 3 \\ e^{x-3} & \text{if } 3 \le x \le 9 \end{cases}$

then $\int_0^9 f(x) dx =$

1) $3-\frac{\pi}{2}-e^6$ 2) $3-\frac{\pi}{2}+e^6$

3) $3+\frac{\pi}{2}-e^6$ 4) $3+\frac{\pi}{2}+e^6$

PROBLEMS BASED ON PROPERTIES

14. $\int_0^1 \frac{\log x}{\sqrt{1-x^2}}dx =$

1) $\pi \log 2$ 2) $-\pi \log 2$

3) $2^{-\log 2}$ 4) $-\frac{\pi}{2}\log 2$

15. $\int_0^{\infty} \frac{\log(1+x^2)}{1+x^2}dx =$

1) $\pi \log 2$ 2) $-\pi \log 2$

3) $-\frac{\pi}{2}\log 2$ 4) $\frac{\pi}{2}\log 2$

16. $\int_{-\pi/4}^{\pi/4} \log(\cos x + \sin x)dx =$

1) $\pi \log 2$ 2) $-\pi \log 2$

3) $-\frac{\pi}{4}\log 2$ 4) $\pi^2 \log 2$

17. $\int_0^{\pi/2} \frac{\sin^2 x}{\sin x + \cos x}dx =$

1) $\sqrt{2}\log(\sqrt{2}+1)$ 2) $\frac{1}{\sqrt{2}}\log(\sqrt{2}+1)$

3) $\log(\sqrt{2}+1)$ 4) $\frac{1}{\sqrt{2}}\log(\sqrt{2}-1)$

18. $\int_0^1 \frac{\mathrm{Sin}^{-1}x}{x}dx =$

1) $\pi \log 2$ 2) $-\pi \log 2$

3) $-\frac{\pi}{2}\log 2$ 4) $\frac{\pi}{2}\log 2$

19. $\int_{2-\sqrt{3}}^{2+\sqrt{3}} \frac{x\,dx}{(1+x)(1+x^2)} = =$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{6}$

3) $\frac{\pi}{12}$ 4) $\frac{\pi}{24}$

20. $\int_0^1 \tan^{-1}\left[\frac{2x-1}{1+x-x^2}\right]dx =$

1\. 0 2. 1/2

3\. 1 4. $\pi/6$

21. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}dx, a>0,$ is

1) 2π 2) $\frac{\pi}{a}$ 3) $\frac{\pi}{2}$ 4) $a\pi$

22. $\int_2^5 \frac{[x^2+49-14x]}{[x^2-14x+49]+[x^2]}dx =$

(where [ ] is g.i.f.)

1) $\frac{1}{2}$ 2) $\frac{3}{2}$ 3) $\frac{5}{2}$ 4) $\frac{7}{2}$

23. Let $\frac{d}{dx}F(x) = \frac{e^{\sin x}}{x}, x>0.$

If $\int_1^4 \frac{3}{x}e^{\sin x^3}dx = F(k)-F(1)$ then one of the possible valuesof k is

1)16 2)63

3)64 4)15

24. Value of $\int_2^3 \frac{dx}{\sqrt{1+x^3}}$ is

1) less than 1
2) greater than 2
3) lies between 3 and 4
4) 0

PROBLEMS BASED ON $\int xf(x)$

25. $\int_{-\pi}^{\pi} \frac{2x(1+\sin x)}{1+\cos^2 x}dx =$

1) $\frac{\pi^2}{4}$ 2) π^2 3) 0 4) $\frac{\pi}{2}$

26. $\int_{-1}^{3}\left(\tan^{-1}\frac{x}{x^2+1}+\tan^{-1}\frac{x^2+1}{x}\right)dx =$

1) π 2) 2π 3) 4π 4) 3π

STEP FUNCTIONS

27. $\int_0^{\pi}[\cot x]dx,\ where[.]$ denotes the greatest integer function, is equal to (AIEEE 2009)

1) 1 2) -1 3) $-\frac{\pi}{2}$ 4) $\frac{\pi}{2}$

28. $\int_0^{1.5} x^2\left[x^2\right]dx =$ (where [ ] is g.i.f.)

1) $\frac{5.75+2\sqrt{2}}{3}$ 2) $\frac{3.375-2\sqrt{2}}{3}$

3) $\frac{4.375-2\sqrt{2}}{3}$ 4) $\frac{5.75-2\sqrt{2}}{3}$

29. $\int_0^{102}\left[\tan^{-1}x\right]dx =$ (where $[x]$ is the largest integer not exceeding x)

1) $102-\tan 1$ 2) 101

3) 102 + tan 1 4) $102-\frac{\pi}{4}$

30. $\int_1^3 \frac{dx}{x^2+[x]^2+1-2x[x]} =$ (where $[x]$ is the largest integer not exceeding x)

1) π 2) 3π 3) 2π 4) $\frac{\pi}{2}$

31. The value of $\int_0^9\left[\sqrt{x}+2\right]dx,$ where $[.]$ is the greatest integer function

1) 31 2) 23 3) 22 4) 27

32 The value of $\int_0^{11}[x]^3\,dx$ where $[.]$ denotes the greatest integer function, is

1) 0 2) 14400

3) 2200 4) 3025

33. $\int_0^{\frac{\pi}{4}} Tan^n\left(x-[x]\right)+Tan^{n-2}\left(x-[x]\right)dx =$ where [ ] is g.i.f

1) $\frac{1}{n}$ 2) $\frac{1}{n-1}$ 3) $\frac{1}{n-2}$ 4) $\frac{1}{n+1}$

34. $\int_{-1/7}^{20/7} e^{5\{x\}}\,dx$ where { } = fractional part of x

1) $3/5\left(e^5\right)$ 2) $\frac{3}{5}\left(e^5-1\right)$

3) 3/5 4) 0

35. If [ ] denotes the greatest integer function then $\int_0^{[x]}\frac{2^x}{2^{[x]}}dx =$

1) $\frac{[x]}{\log 2}$ 2) $\frac{-[x]}{\log 2}$ 3) $\frac{-[2x]}{\log 2}$ 4) $\frac{[2x]}{\log 2}$

REDUCTION FORMULA

36. $\int_0^{\pi/4}\sin^4 x.\cos^4 x dx =$

1) $\frac{3\pi}{512}$ 2) $\frac{\pi}{512}$ 3) $\frac{-\pi^2}{512}$ 4) $\frac{7\pi}{512}$

37. $I_n=\int_1^e(\log x)^n dx$ and $I_n=A+BI_{n-1}$ then A=........, B=......

1) e, -n 2) 1/e, n 3) -e, n 4) -e ,-n

38. If $I_n=\int_0^{\frac{\pi}{4}}\tan^n x\,dx,$ then $\lim_{n\to\infty} n\left[I_n+I_{n+2}\right]=$

1) $\frac{1}{2}$ 2) 1

3) ∞ 4) 0

39. If $g(x)=\int_0^x\cos 4t\,dt$, then $g(x+\pi)$ equals [AIEEE 2012]

1) $\frac{g(x)}{g(\pi)}$ 2) $g(x)\pm g(\pi)$

3) $g(\pi)/g(x)$ 4) $g(x).g(\pi)$

PROBLEMS BASED ON LEIBNITZ RULE

40. $\lim\limits_{x\to\infty}\dfrac{\int_0^x\left(\tan^{-1}t\right)^2 dt}{\sqrt{x^2+1}}=$

1) $\dfrac{\pi}{4}$ 2) $\dfrac{\pi^2}{2}$ 3) $\dfrac{\pi^2}{4}$ 4) π

41. The value of $\underset{x\to 0}{Lt}\dfrac{\left(\int_0^x e^x dx\right)^2}{\int_0^x e^{2x^2}dx}$ **is**

1) 1 2) 2 3) 3 4) 0

MISCELLANEOUS PROBLEMS

42. If $\int_e^x tf(t)dt=\sin x-x\cos x-\dfrac{x^2}{2}$ **for all** $x\in R-\{0\}$ **then the value of** $f(\pi/6)$ **is**

1) 0 2) 1 3) $-1/2$ 4) $3/2$

43. If $\int_{\frac{\pi}{3}}^{x}\sqrt{3-\sin^2 t}\,dt+\int_0^y \cos t\,dt=0$ **then** $\dfrac{dy}{dx}=$

1) $\dfrac{\sqrt{3-\sin^2 x}}{\cos y}$ 2) $-\dfrac{\sqrt{3-\sin^2 x}}{\cos y}$ 3)

$\cos y\sqrt{3-\sin^2 x}$ 4) $-\cos y\sqrt{3-\sin^2 x}$

44. If $\int_0^1\dfrac{e^t dt}{t+1}=a$ **then** $\int_{b-1}^{b}\dfrac{e^{-t}dt}{t-b-1}$ **is equal to**

1) ae^{-b} 2) $-ae^{-b}$

3) $-be^{-a}$ 4) ae^{b}

45. If I = $\int_0^1\sqrt{1+x^3}$ **dx then**

1) $I>1$ 2) $I>\dfrac{\sqrt{5}}{2}$

3) $I>\dfrac{\sqrt{7}}{2}$ 4) $I<1$

46. If $I_1=\int_0^1\dfrac{1}{|x|}dx$ *and* $I_2=\int_0^1\dfrac{1}{\sqrt{1+x^2}}dx$ **then**

1) $I_1=I_2$ 2) $I_1<I_2$

3) $I_1>I_2$ 4) $I_1=5I_2$

47. If $I_1=\int_0^1 2^{x^2}dx$, $I_2=\int_0^1 2^{x^3}dx$, $I_3=\int_1^2 2^{x^2}$ **dx and** $I_4=\int_1^2 2^{x^3}dx$, **then**

1) $I_1>I_2$ 2) $I_2>I_1$ 3) $I_3>I_4$ 4) $I_1>I_3$

48. If $I_1=\int_0^{\pi/2}\cos(\sin x)dx$; $I_2=\int_0^{\pi/2}\sin(\cos x)dx$ **and** $I_3=\int_0^{\pi/2}\cos x dx$, **then**

1) $I_1>I_3>I_2$ 2) $I_3>I_1>I_2$

3) $I_1>I_2>I_3$ 4) $I_3>I_2>I_1$

49. $I_1=\int_0^1\left(1-x^{50}\right)^{101}dx,\ I_2=\int_0^1\left(1-x^{50}\right)^{100}dx$ **then**

1) $51I_1=50I_2$ 2) $101I_1=100I_2$

3) $5051I_1=5050I_2$ 4) $5050I_1=5051I_2$

50. If $f(x)=\int_{-1}^{x}t\left(e^t-1\right)(t-1)(t-2)^3(t-3)^5$ **then** $f^1(1)$ **equals**

1) 1 2) 2 3) 0 4) 4

51. If $K\le\int_0^{2\pi}\dfrac{dx}{5+4\cos x}\le l$ *then* $[k,l]=$

1) $[0\ 2\pi]$ 2) $\left[\dfrac{2\pi}{5}\ \dfrac{2\pi}{4}\right]$

3) $\left[0\ \dfrac{2\pi}{9}\right]$ 4) $\left[\dfrac{2\pi}{9}\ 2\pi\right]$

52. $\int_0^1\sqrt{(1+x)(1+x^3)}\,dx$ **can not exceed**

1) $\sqrt{\dfrac{15}{8}}$ 2) $\sqrt{\dfrac{13}{8}}$ 3) $\sqrt{\dfrac{11}{8}}$ 4) $\sqrt{\dfrac{9}{8}}$

PROBLEMS BASED ON DEFINITE INTEGRATION AS LIMIT OF SUM

53. $\underset{n\to\infty}{Lt}\left[\frac{1}{n^2}\sec^2\frac{1}{n^2}+\frac{2}{n^2}\sec^2\frac{4}{n^2}+...+\frac{1}{n}\sec^2 1\right]$

1) $\frac{1}{2}\sec 1$ 2) $\frac{1}{2}co\sec 1$ 3) $\tan 1$ 4) $\frac{1}{2}\tan 1$

54. $\lim_{n\to\infty}\frac{(n!)^{1/n}}{n}$ **is equal to**

1) e 2) $\frac{1}{e}$ 3) $e-1$ 4) e - 5

55. If $S_n=\left[\frac{1}{1+\sqrt{n}}+\frac{1}{2+\sqrt{2n}}+....+\frac{1}{n+\sqrt{n^2}}\right]$ **then**

$\lim_{n\to\infty}S_n=$

1) $\log 2$ 2) $\log 4$ 3) $\log 8$ 4)$\log 5$

56. Evaluate

$\underset{n\to\infty}{Lim}\left[\frac{1}{\sqrt{4n^2-1}}+\frac{1}{\sqrt{4n^2-4}}+\frac{1}{\sqrt{4n^2-9}}+.....+\frac{1}{\sqrt{3n^2}}\right]$

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) π

LEVEL-II (C.W) - KEY

1) 2	2) 2	3) 3	4) 3	5) 1	6) 3
7) 2	8) 4	9) 4	10) 4	11) 3	12) 2
13) 2	14) 4	15) 1	16) 3	17) 2	18) 4
19) 2	20) 1	21) 3	22) 2	23) 3	24) 1
25) 2	26) 1	27) 3	28) 4	29) 1	30) 4
31) 1	32) 4	33) 2	34) 2	35) 1	36) 1
37) 1	38) 2	39) 2	40) 3	41) 4	42) 3
43) 2	44) 2	45) 1	46) 3	47) 1	48) 1
49) 3	50) 3	51) 4	52) 1	53) 4	54) 2
55) 2	56) 1				

LEVEL-II (C.W) - HINTS

1. $\frac{d}{dx}\left(\tan^{-1}\left(\frac{1}{x}\right)\right)=\frac{-1}{1+x^2}$

2. $\int_o^1\sqrt{\frac{x}{1-x^3}}dx=\int_o^1\frac{\sqrt{x}}{\sqrt{1-\left(x^{\frac{3}{2}}\right)^2}}dx$

3. Use $\int\frac{1}{(x^2+a^2)(x^2+b^2)}dx=$

$\frac{1}{b^2-a^2}\left[\frac{1}{a}\tan^{-1}\frac{x}{a}-\frac{1}{b}\tan^{-1}\frac{x}{b}\right]$

4. $\int_1^{\sqrt[7]{2}}\frac{1}{x(2x^7+1)}dx$

$=\frac{1}{7}[\log y-\log(2y+1)]_1^2=\frac{1}{7}\left[\log\frac{2}{5}-\log\frac{1}{3}\right]$

$=\frac{1}{7}\log\left(\frac{6}{5}\right)$

5. $x=\cos 2\theta$

6. $\int_{1/3}^{1}\frac{(x-x^3)^{1/3}}{x^4}dx=\int_{1/3}^{1}\cancel{x}\frac{\left(\frac{1}{x^2}-1\right)^{1/3}}{x^{4^3}}dx$ Put $\frac{1}{x^2}-1=t$

7. using the formula $\int_0^{\frac{\pi}{2}}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}=\frac{\pi}{2ab}$

8. $Tan\,x/2=t$

9. Put $x=\tan\theta\Rightarrow dx=\sec^2\theta\,d\theta$

$\therefore\int_0^1\frac{\tan^{-1}x}{x}dx=\int_0^{\frac{\pi}{4}}\frac{\theta}{\tan\theta}\sec^2\theta\,d\theta$

$2\int_0^{\frac{\pi}{4}}\frac{\theta}{\sin 2\theta}d\theta$

$\frac{1}{2}\int_0^{\frac{\pi}{2}}\frac{t}{\sin t}dt$ $where\ 2\theta=t$ Hence $k=\frac{1}{2}$

10. $\sqrt{x+\left(2\sqrt{x-1}\right)}=\left|\sqrt{x-1}+1\right|$ and

$\sqrt{x-2\sqrt{x-1}}=\left|\sqrt{x-1}-1\right|$

$I=\int_1^2 2dx+\int_2^5 2\sqrt{x-1}dx$

11. $I=\int_{-2}^{0}\left[(x+1)^3+2+(x+1)\cos(x+1)\right]dx$

Put $x+1=t$ $\therefore I=\int_{-1}^{1}\left(t^3+2+t\cos t\right)dt$

12. $put\sin^{-1}x=t$
and use byparts

13. $I=\int_{0}^{\pi/2}\sin x\,dx+\int_{\pi/2}^{3}1\,dx+\int_{3}^{9}e^{x-3}dx$

14. put $x=\sin\theta$

15. Put $x=\tan\theta$

16. Using formula $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$

17. Using formula $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

18. $x=\sin\theta$

19. Put $x=\tan\theta$

20. $Tan^{-1}\left(\frac{2x-1}{1+x-x^2}\right)=Tan^{-1}\left(\frac{x-(1-x)}{1+x(1-x)}\right)$

$\tan^{-1}x-\tan^{-1}(1-x)$

21. $I=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx$ _____(1)

Apply $f(a+b-x)$ formula.

$=\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^{-x}}dx$ _____(2) adding (1) and(2)

22 38. Apply $f(a+b-x)$ formula.

$I=\int_2^5\frac{\left[(7-x)^2\right]}{\left[(7-x)^2\right]+\left[x^2\right]}dx$

23. $\frac{d}{dx}(f(x))=\frac{e^{\sin x}}{x}\Rightarrow F(x)=\int\frac{e^{\sin x}}{x}dx$

$F(k)-F(1)=\int_1^4\frac{3}{x}e^{\sin x^3}dx$ [put $x^3=t$]

$=\int_1^{64}\frac{e^{\sin t}}{t}dt$ $=\left[F(t)\right]_1^{64}$

$=F(64)-F(1)$ $\therefore K=64$

24 $1+x^3>(1+x)^2$ in $x\in(2,3)$

$\sqrt{1+x^3}>(1+x)$ $\Rightarrow\int_2^3\frac{1}{\sqrt{1+x^3}}dx<\int_2^3\frac{1}{1+x}dx$

$I<\log_e\left(\frac{4}{3}\right)\Rightarrow I<1$

25. Given$=\int_{-\pi}^{\pi}\frac{2x}{1+\cos^2 x}dx+\int_{-\pi}^{\pi}\frac{2x\sin x}{1+\cos^2 x}dx$

$=4\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$

$\because$ $\frac{2x}{1+\cos^2 x}$ is odd and $\frac{x\sin x}{1+\cos^2 x}$ is even apply $\int_0^{2a}f(x)dx=2\int_0^{a}f(x)dx$ if $f(2a-x)=f(x)$ and put $\cos x=t$

26. $\tan^{-1}x=-\pi+\cot^{-1}\frac{1}{x}$ If $x<0$

$\tan^{-1}x=\cot^{-1}\frac{1}{x}$ If $x>0$

27. Apply $f(a+b-x)$ formula.

$I=\int_0^{\pi}[\cot x]dx,\ I=\int_0^{\pi}[-\cot x]dx$

$I+I=\int_0^{\pi}[\cot x]dx+\int_0^{\pi}[-\cot x]dx=\int_0^{\pi}-1dx$

28. $I=\int_0^1 0dx+\int_1^{\sqrt2}x^2dx+\int_{\sqrt2}^{1.5}2x^2dx$

$=\left(\frac{x^3}{3}\right)_1^{\sqrt2}+\left(\frac{2x^3}{3}\right)_{\sqrt2}^{1.5}=\frac{2\sqrt2-1}{3}+\frac{2}{3}\left(3.375-2\sqrt2\right)$

29. for $0\le x\le\tan 1$,
$0\le\tan^{-1}x<1$ and $\tan 1\le x\le 102$

$1\le\tan^{-1}<2$; $\int_0^{102}\left[\tan^{-1}x\right]dx=\int_0^{\tan 1}odx+\int_{\tan 1}^{102}1dx$

$=(x)_{\tan 1}^{102}=102-\tan 1$

30. $I=\int_1^3 \frac{dx}{(x-[x])^2+1}=\int_1^2 \frac{dx}{(x-[x])^2+1}+\int_2^3 \frac{dx}{(x-[x])^2+1}$

$=\int_1^2 \frac{dx}{(x-1)^2+1}+\int_2^3 \frac{dx}{(x-2)^2+1}$

$=\left(\tan^{-1}(x-1)\right)_1^2+\left(\tan^{-1}(x-2)\right)_2^3$

31. $\int_0^9 \left[\sqrt{x}+2\right]dx$

$=\int_0^1\left[\sqrt{x}+2\right]dx+\int_1^4\left[\sqrt{x}+2\right]dx+\int_4^9\left[\sqrt{x}+2\right]dx$

$=\int_0^1 2dx+\int_1^4 (1+2)\,dx+\int_4^9 (2+2)\,dx$

32. $\int_0^{11}[x]^3\,dx=\sum_{n=0}^{10}\int_r^{r+1}[x]^3\,dx=\sum_{n=0}^{10}\int_n^{r+1} r^3dx$

$=\sum_{r=0}^{10}\left(r^3\right)=0^3+1^3+2^3+...+10^3=\left(\frac{10\times 11}{2}\right)^2=3025$

33. $I=\int_0^{\pi/4}\tan^n x+\tan^{n-2}x\;dx$

$=\int_0^{\pi/4}\tan^{n-2}x\sec^2 x\;dx$

34. $\int_{-1/7}^{3-1/7} e^{5\{x\}}\,dx=3\int_0^1 e^{5\{x\}}\,dx=3\left[\frac{e^5-1}{5}\right]$

35. $\int_0^{[x]} 2^{x-[x]}=[x]\int_0^1 2^{x-[x]}\;dx=[x]\int_0^1 2^x\;dx$

36. Using reduction formula
37. Using by parts
38. Using reduction formula

39. $g(x)=\int_0^x \cos 4t\,dt=\left(\frac{\sin 4t}{4}\right)_0^x=\frac{\sin 4x}{4}$

$g(x+\pi)=\int_0^{x+\pi}\cos 4t\,dt=\left[\frac{\sin(4t)}{4}\right]_0^{x+\pi}$

$=\frac{1}{4}\sin 4(\pi+x)=\frac{1}{4}\sin(4x)$

$g(\pi)=\int_0^{\pi}\cos 4t\,dt=0$;

$g(x+\pi)=g(x)+g(\pi)$

$g(x+\pi)=g(x)-g(\pi)$

40. Using Leibnitz rule
41. Using Leibnitz rule
42. **Solution :** Differentiating both the sides and $xf(x)=\cos x-\cos x+x\sin x-x$, so $f(x)=\sin x-1$. Hence $f(\pi/6)=\sin\pi/6-1=-1/2$

43. Differentiating the given equation with respect to x,

we get $\frac{d}{dx}\left[\int_{\frac{\pi}{3}}^{x}\sqrt{3-\sin^2 x}\right]+\frac{d}{dx}\left[\int_0^y \cos t dt\right]=0$

$\Rightarrow\sqrt{3-\sin^2 x}.\frac{d}{dx}x-\sqrt{3-\sin^2\frac{\pi}{3}}\frac{d}{dx}\left(\frac{\pi}{3}\right)$
$+\cos y\frac{dy}{dx}-\cos(0)\frac{d(0)}{dx}=0$

$\Rightarrow\sqrt{3-\sin^2 x}-0+\cos y\frac{dy}{dx}-0=0\Rightarrow\frac{dy}{dx}=-\frac{\sqrt{3-\sin^2 x}}{\cos y}$

44. $\because a=\int_0^1\frac{e^t}{t+1}dt$

$\therefore\int_{b-1}^{b}\frac{e^{-t}dt}{t-b-1}=-e^{-b}\int_0^1\frac{e^y}{y+1}dy$

Put $\begin{matrix} t=b-y \\ dt=-dy \end{matrix}$ $=-e^{-b}.a$

45. $o<x<1\Rightarrow 1<\sqrt{1+x^3}<\sqrt{2}$

$\Rightarrow\int_0^1 1dx<\int_0^1\sqrt{1+x^3}dx$

46. since $|x|<\sqrt{1+x^2}$

$\therefore\frac{1}{|x|}>\frac{1}{\sqrt{1+x^2}}$

$\Rightarrow\int_0^1\frac{1}{|x|}dx>\int_0^1\frac{1}{\sqrt{1+x^2}}dx\Rightarrow I_1>I_2$

47. $0<x<1 \Rightarrow x^2 > x^3 \Rightarrow 2^{x^2} > 2^{x^3}$

$\therefore \int_1^2 f(x)\,dx = \int_1^2 \left(\frac{-2}{5x}+\frac{3x}{5}-\frac{2}{5}\right)dx$

48. we have, $\sin x < x$ for $x>0$

$\Rightarrow \sin(\cos x) < \cos x$ for $0<x<\pi/2$

$\Rightarrow \int_0^{\pi/2} \sin(\cos x)\,dx < \int_0^{\pi/2} \cos x\,dx \qquad \therefore I_3 > I_2$

Now $\cos x < \cos\alpha$ if $x>\alpha$ and x, $\alpha \in [0, \frac{\pi}{2}]$

$\therefore \; x > \sin x \quad \Rightarrow \cos x < \cos(\sin x)$

$\Rightarrow \int_0^{\pi/2} \cos x dx < \int_0^{\pi/2} \cos(\sin x)\,dx \qquad \therefore I_3 < I_1$

$\therefore$ from (1) and (2) $I_1 > I_3 > I_2$

49. $I_1 = \left\langle \left(1-x^{50}\right)^{101} x \right\rangle_0^1 - \int_0^1 x101\left(1-x^{50}\right)^{100}\left(-50x^{49}\right)dx$

$= (101)(50)\int_0^1 x^{50}\left(1-x^{50}\right)^{100}$

50 $f^1(x) = x\left(e^x-1\right)(x-1)(x-2)^3(x-3)^5$

51. $-1 \le \cos x \le 1 \Rightarrow \frac{1}{9} \le \frac{1}{5+4\cos x} \le 1$

apply integration with given limits

52. $I \le \sqrt{\int_0^1 (1+x)\,dx \int_0^1 \left(1+x^3\right)dx}$

53. $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{r}{n^2}\sec^2\left(\frac{r}{n}\right)^2 = \int_0^1 x\sec^2 x^2 dx$

54. Let $y = \lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \lim_{n\to\infty}\left(\frac{n!}{n^n}\right)^{1/n}$

$\Rightarrow \log y = \lim_{n\to\infty} \frac{1}{n}.\log\left[\frac{1}{n}.\frac{2}{n}.\frac{3}{n}......\frac{n}{n}\right]$

$= \lim_{n\to\infty} \frac{1}{n}\left[\log\frac{1}{n}+\log\frac{2}{n}+....\log\frac{n}{n}\right]$

$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}\log\frac{r}{n} = \int_0^1 \log x dx$

55. $\lim_{n\to\infty} S_n = \lim_{n\to\infty}\left[\frac{1}{1+\sqrt{n}}+\frac{1}{2+\sqrt{2n}}+....+\frac{1}{n+\sqrt{n^2}}\right]$

$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{r+\sqrt{rn}} = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n}.\frac{1}{\frac{r}{n}+\sqrt{\frac{r}{n}}}$

$= \int_0^1 \frac{dx}{x+\sqrt{x}}$

56. $L = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{\sqrt{4n^2-r^2}} \Rightarrow L = \int_0^1 \frac{dx}{\sqrt{4-x^2}}$

LEVEL-II (H.W)

PROBLEMS BASED ON BASIC FORMULA AND SUBSTITUTIONS

1. $\int_1^4 \frac{xdx}{\sqrt{2+4x}} =$

1) $\frac{1}{2}$ 2) $\frac{1}{\sqrt{2}}$ 3) $\frac{3}{2}$ 4) $\frac{3}{\sqrt{2}}$

2. $\int_0^1 \frac{x}{1+\sqrt{x}}dx =$

1) $\frac{5}{3}$ - log 4 2) $\frac{5}{3}$ + log 4

3) $\frac{5}{3}$ log 4 4) $\frac{3}{5}$ – log 4

3. $\int_{\log 2}^{t} \frac{dx}{\sqrt{e^x-1}} = \frac{\pi}{6}$**, then t=**

1) 4 2) log 8 3) log 4 4) log2

4. $\int_0^{\pi/4} \frac{\sin x+\cos x}{3+\sin 2x}$ **dx is equal to**

1) $-\frac{1}{4}\log 3$ 2) $\frac{1}{4}\log 3$

3) $-\frac{1}{3}\log 4$ 4) $\frac{1}{2}\log 3$

5. $\int_0^{\pi} \frac{dx}{3+2\sin x+\cos x} =$

1) $\pi/3$ 2) $\pi/4$ 3) $\pi/6$ 4) $\pi/2$

6. $\int_0^{2\pi} \sqrt{1+\sin\frac{x}{2}}\, dx =$

1)0 2)2 3)8 4)4

7. $\int_0^1 \frac{dx}{x^2+2x\cos\alpha+1} =$

1) $\sin\alpha$ 2) $\tan^{-1}(\sin\alpha)$

3) $\frac{\alpha}{2\sin\alpha}$ 4) $\alpha\sin\alpha$

8. **The value of** $\int_0^{\pi/2} \frac{x+\sin x}{1+\cos x}$ **dx is**

1) π 2) 2π 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{4}$

9 $\int_0^{\pi/2} \sqrt{\cos x}\sin^5 x\, dx =$

1) $\frac{34}{231}$ 2) $\frac{64}{231}$

3) $\frac{30}{321}$ 4) $\frac{128}{231}$

10. $\int_0^1 \sin^{-1}\left(\frac{2x}{1+x^2}\right) dx =$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{4}+\log 2$

3) $\frac{\pi}{2}+\frac{1}{2}\log 2$ 4) $\frac{\pi}{2}-\log 2$

11. **The value of** $\int_0^1 \frac{8\log(1+x)}{1+x^2} dx$ **is**

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1) $\pi\log 2$ 2) $\frac{\pi}{8}\log 2$

3) $\frac{\pi}{2}\log 2$ 4) $\log 2$

PROBLEMS BASED ON PROPERTIES

12. **If** $\int_0^{\pi/2} \log(\sin x)dx = k$ **then** $\int_0^{\pi/4} \log(1+\tan x)dx =$

1) $\frac{-k}{8}$ 2) $\frac{-k}{4}$ 3) $\frac{k}{8}$ 4) $\frac{k}{4}$

13. **If f(x) = f(a-x) and g(x) + g(a-x)=2 then**

$\int_0^a f(x).g(x)dx =$

1) 0 2) a

3) $\int_0^a f(x)dx$ 4) a^2

14. $\int_0^4 \frac{1}{1+x^2} dx = =$

1) $\pi\log 2$ 2) $\tan^{-1}(4)$

3) $\frac{\pi}{4}\log 2$ 4) 0

15. $\int_0^{2\pi} \frac{1}{1+\tan^4 x} dx =$

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{3\pi}{4}$ 4) π

16. $\int_0^1 \cot^{-1}(1-x+x^2)dx =$

1) $\frac{\pi}{2}$ - log 2 2) log 2

3) $\frac{\pi}{2}+\frac{1}{2}\log 2$ 4) $\frac{\pi}{2}$

17. $\int_0^a \frac{dx}{x+\sqrt{a^2-x^2}} =$

1) π 2) $\pi/3$

3) $-\pi$ 4) $\pi/4$

PROBLEMS BASED ON $\int xf(x)$

18. $\int_0^{\pi} \frac{x dx}{1+\cos^2 x} =$

1) 0 2) $\frac{\pi^2}{4}$ 3) $\frac{\pi^2}{2\sqrt{2}}$ 4) $\frac{\pi^2}{8}$

19. $\int_0^{\pi} x.\log(\sin x)dx =$

1) $\pi^2 \log 2$ 2) $\frac{\pi^2}{2}\log 2$

3) $\frac{\pi^2}{4}$ 4) $\frac{-\pi^2}{2}\log 2$

STEP FUNCTIONS

20. $\int_0^4 [2x+3]dx =$

1) 12 2) 24 3) 26 4) 0

21. $\int_0^{\pi/3} \left[\sqrt{3}\tan x\right]$dx= (where [ ] is g.i.f.)

1) $\frac{5\pi}{6}$ 2) $\frac{5\pi}{6} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

3) $\frac{\pi}{2} - \tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ 4) $\frac{\pi}{2}$

22. The value of $\int_1^a [x] f'(x)dx,\ a>1$, where $[x]$ denotes the greatest integer not exceeding x is

1) $[a]f([a]) - \{f(1)+f(2)+......+f(a)\}$

2) $af([a]) - \{f(1)+f(2)+......+f(a)\}$

3) $a\,f(a) - \{f(1)+f(2)+......+f(a)\}$

4) $[a]f(a) - \{f(1)+f(2)+......+f[a]\}$

23. $\int_1^{\infty}\left[\frac{1}{1+x^2}\right]dx =$ where [ ] $g.i.f$

1) 0 2) ∞ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{2}$

24. $\int_0^{\infty}\left[\frac{2}{e^x}\right]dx$, where [ ] denotes the greatest integer function.

1) $\log_e 3$ 2) $\log_e 2$ 3) $\log_e 5$ 4) 0

25. $\int_{-20\pi}^{20\pi} |\sin x|[\sin x]dx =$ where [ ] $g.i.f$

1) - 20 2) - 40 3) - 30 4) - 60

26. $\int_0^{64}\left\{\sqrt{x}\right\} =$ { } = fractional part of x

1) 90 2) $\frac{100}{3}$ 3) 70 4) 60

27. $\int_0^{64}\left\{\sqrt[3]{x}\right\}dx =$ { } = fractional part of x

1) 24 2) 36 3) 12 4) 48

28. if $[x]$ stands the greatest integer function

$\int_{0.5}^{4.5}[x]\,dx + \int_{-1}^{1}|x|\,dx =$

1) 9 2) 8 3) 7 4) 6

29. $\int_{-3}^{3} x^8\{x^{11}\}dx =$ { } = fractional part of x

1) 3^6 2) 3^7 3) 3^8 4) 3^9.

REDUCTION FORMULA

30. $\int_0^3 (9-x^2)^{3/2}dx =$

1) $\frac{243\pi}{16}$ 2) $\frac{\pi}{16}$ 3) $\frac{-243\pi^2}{15}$ 4) $\frac{81\pi}{16}$

31. $\int_0^{\pi/4} \tan^4 x dx =$

1) $\frac{\pi}{4}-\frac{2}{3}$ 2) $\frac{\pi}{4}+\frac{2}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi^2}{4}+\frac{3}{2}$

32. $\int_{-\pi/2}^{\pi/2} \sin^2 x.\cos^2 x(\sin x+\cos x)dx =$

1) $\frac{1}{15}$ 2) $\frac{2}{15}$ 3) $\frac{4}{15}$ 4) 0

33. $\int_0^{2\pi} x.\sin^4 x.\cos^6 x.dx =$

1) $\frac{3\pi^2}{128}$ 2) $\frac{15\pi^2}{128}$ 3) $\frac{3\pi^2}{64}$ 4) $\frac{5\pi^2}{128}$

34. If $f(x)=\int_0^x (\sin^4 t+\cos^4 t)dt$ then $f(x+\pi)$ is equal to

1) $f(x)+f\left(\frac{\pi}{2}\right)$ 2) $f(x)+2f\left(\frac{\pi}{2}\right)$

3) $f(x)-f\left(\frac{\pi}{2}\right)$ 4) $f(x)-2f\left(\frac{\pi}{2}\right)$

MISCELLANEOUS PROBLEMS

35. Equation of the tangent line to the curve $y=\int_0^x \cos\theta d\theta$ at $x=\frac{\pi}{2}$ is

1) $x=0$ 2) $y=0$ 3) $x=1$ 4) $y=1$

36. Let f be integrable over $[0,a]$ for any real a. If we define

$I_1=\int_0^{\pi/2} \cos\theta.f(\sin\theta+\cos^2\theta)d\theta$ and

$I_2=\int_0^{\pi/2} \sin 2\theta\, f(\sin\theta+\cos^2\theta)d\theta$ then

1) $I_1=I_2$ 2) $I_1=-I_2$

3) $I_1=2I_2$ 4) $I_1=-2I_2$

37. Let $f:R\to R$ be a continuous function given by $f(x+y)=f(x)+f(y)$ for all $x,y\in R$. If $\int_0^2 f(x)dx=\alpha$, then $\int_{-2}^2 f(x)dx$ is equal to

1) 2α 2) α 3) 0 4) $\alpha-2$

38. The natural number $n(\le 5)$ for which $I_n=\int_0^1 e^x(x-1)^n dx=16-6e$ is

1) 2 2) 3 3) 4 4) 5

39 $\int_3^4 \frac{dx}{\sqrt[3]{\ln x}}$

1) $<\frac{1}{2}$ 2) $>\frac{1}{2}$ 3) $=\frac{1}{2}$ 4) <0

40. If $I_{(m,n)}=\int_0^1 t^m(1+t)^n dt$, then the expression for $I_{(m,n)}$ in terms of $I_{(m+1,n-1)}$ is

1) $\frac{2^n}{m+1}-\frac{n}{m+1}I(m+1,n-1)$

2) $\frac{n}{m+1}I(m+1,n-1)$

3) $\frac{2^n}{m+1}+\frac{n}{m+1}I(m+1,n-1)$

4) $\frac{m}{m+1}I(m+1,n-1)$

41. $\left(\sum_{n=1}^{10}\int_{-2n-1}^{-2n}\sin^{27}xdx\right)+\left(\sum_{n=1}^{10}\int_{2n}^{2n+1}\sin^{27}xdx\right)=$

1) 27^2 2) -54 3) 54 4) 0

42. If f(x)=$\begin{vmatrix}\sin x+\sin 2x+\sin 3x & \sin 2x & \sin 3x\\ 3+4\sin x & 3 & 4\sin x\\ 1+\sin x & \sin x & 1\end{vmatrix}$ then the value of $\int_0^{\pi/2} f(x)dx$ is

1) 3 2) 2/3 3) 1/3 4) 0

43. The integral $\int_{\pi/4}^{5\pi/4}\left(|\cos t|\sin t+|\sin t|\cos t\right)dt$ has the value equal to

1) 0 2) 1/2

3) $1/\sqrt{2}$ 4) 1

44. $I=\int_0^1 e^{x^2}$ satisfies inequality

1) $o<I<1$ 2) $-1\le I\le 1$

3) $1\le I\le e$ 4) $-1\le I\le 0$

45. $I=\int_0^{\frac{1}{2}}\frac{1}{\sqrt{1-x^{2n}}}dx$ $n\ge 1$ then

1) I$>\frac{1}{2}$ 2) I$<\frac{1}{4}$

3) I$<\frac{1}{3}$ 4) I>1

PROBLEMS BASED ON DEFINITE INTEGRATION AS LIMIT OF SUM

46. $\underset{n\to\infty}{Lt}\dfrac{1}{\sqrt{2n-1^2}}+\dfrac{1}{\sqrt{4n-2^2}}+\dfrac{1}{\sqrt{6n-3^2}}+....+\dfrac{1}{n}$

1) $\dfrac{\pi}{4}$ 2) $\dfrac{\pi}{2}$ 3) $\dfrac{\pi}{6}$ 4) $\dfrac{\pi}{3}$

47. $\underset{n\to\infty}{Lt}\left\{\left(1+\dfrac{1}{n}\right)\left(1+\dfrac{2}{n}\right)....\left(1+\dfrac{n}{n}\right)\right\}^{1/n}=$

1) e 2) 1/e 3) 4/e 4) 8/e

48. $\underset{n\to\infty}{Lt}\left\{\left(1+\dfrac{1}{n^2}\right)\left(1+\dfrac{2^2}{n^2}\right)....\left(1+\dfrac{n^2}{n^2}\right)\right\}^{1/n}=$

1) $e^{(\pi-4)/2}$ 2) $2e^{(\pi-4)/2}$

3) $\dfrac{e^{(\pi-4)/2}}{2}$ 4) $e^{(\pi-4)}$

LEVEL-II (H.W.) - KEY

1) 4	2) 1	3)3	4)2	5)2	6)3
7) 3	8) 3	9) 2	10) 4	11) 1	12) 2
13) 3	14) 2	15) 4	16)1	17) 4	18) 3
19) 4	20) 3	21) 3	22) 4	23) 1	24) 2
25) 2	26)2	27) 2	28) 1	29) 2	30) 1
31) 1	32) 3	33) 1	34) 2	35) 4	36) 1
37) 3	38) 2	39) 2	40) 1	41) 4	42) 3
43) 1	44) 3	45) 1	46) 2	47) 3	48) 2

LEVEL-II (H.W.) - HINTS

1. $2+4x=t^2$ 2. put $\sqrt{x}=t$

3. $e^x-1=t^2$ 4. Put $\sin x-\cos x=t$

5. $Tan\ x/2=t$

6. $I=\int_0^{2\pi}\sqrt{\left(\cos\dfrac{x}{4}+\sin\dfrac{x}{4}\right)^2}dx$

$=\int_0^{2\pi}\left(\cos\dfrac{x}{4}+\sin\dfrac{x}{4}\right)dx=\left[4\sin\dfrac{x}{4}-4\cos\dfrac{x}{4}\right]_o^{2\pi}$

$4\left[\sin\dfrac{\pi}{2}-\cos\dfrac{\pi}{2}-(\sin 0-\cos 0)\right]=8$

7. $x^2+2x\cos\alpha+1=(x+\cos\alpha)^2+\sin^2\alpha$

8. $I=\int_0^{\pi/2}\dfrac{x+\sin x}{1+\cos x}$dx

$=\int\limits_0^{\pi/2}\dfrac{1}{2}x\sec^2(x/2)\,dx+\int\limits_0^{\pi/2}\tan(x/2)\,dx$

$=\left(x\tan\dfrac{x}{2}\right)_0^{\pi/2}-\int_0^{\pi/2}\tan\dfrac{x}{2}dx+$

$\int_0^{\pi/2}\tan\dfrac{x}{2}dx=\dfrac{\pi}{2}$

9. Given$=\int\limits_0^{\pi/2}\sqrt{\cos x}\left(1-\cos^2 x\right)^2.\sin x dx$

Put $\cos x=t$

10. $\sin^{-1}\left(\dfrac{2x}{1+x^2}\right)=2\tan^{-1}x$

11. Put $x=\tan\theta$

12. $k=-\dfrac{\pi}{2}\log 2;\ \int\limits_0^{\pi/4}\log(1+\tan x)\,dx=\dfrac{\pi}{8}\log 2$

$\int\limits_0^{\pi/4}\log(1+\tan x)\,dx=\dfrac{\pi}{8}\log 2$

13. $\int\limits_0^a f(x)g(x)dx=\int\limits_0^a f(x)[2-g(a-x)]dx$

14. Put $x=\tan\theta$

15. $\int\limits_a^{2a}f(x)dx=2\int\limits_0^a f(x)dx$ If $f(2a-x)=f(x)$

16. $\tan^{-1}\left(\dfrac{1}{1-x+x^2}\right)=\tan^{-1}\left(\dfrac{x+(1-x)}{1-x(1-x)}\right)$

$=\tan^{-1}x+\tan^{-1}(1-x)$

17. Put $x=a\sin\theta$ and using

18. Using $\int\limits_0^a f(x)dx=\int\limits_0^a f(a-x)dx$

19. Using $\int\limits_0^a f(x)dx=\int\limits_0^a f(a-x)dx$

20. put $2x+3=t$

21. $\int\limits_0^{\frac{\pi}{6}}0dx+\int\limits_{\frac{\pi}{6}}^{\tan^{-1}\frac{2}{\sqrt3}}1dx+\int\limits_{\tan^{-1}\frac{2}{\sqrt3}}^{\frac{\pi}{3}}2\,dx$

22. $I=\int_1^a[x]f^1(x)dx$

$=\int_1^2[x]f^1(x)dx+\int_2^3[x]f^1(x)dx+...+\int_{[a]}^a[x]f^1(x)dx$

$=[a]f(a)-\{f(1)+f(2)+...+f[a]\}$

23. $0<\dfrac{1}{1+x^2}<1$

24. $\int_0^{\ln 2}\left[\dfrac{2}{e^x}\right]dx+\int_{\ln 2}^{\infty}\left[\dfrac{2}{e^x}\right]dx$

25. $I=\int\limits_{-20\pi}^{20\pi}|\sin x|[\sin x]dx$ *then*

$2I=\int\limits_{-20\pi}^{20\pi}\{|\sin x|[\sin x]+[\sin x][-\sin x]\}dx$

$I=-\int\limits_{-20\pi}^{20\pi}|\sin x|dx \because [x]+[-x]=-1$

26. $I=\int\limits_0^{64}\left(\sqrt{x}-\left[\sqrt{x}\right]\right)dx$

$=\int\limits_0^{64}\sqrt{x}\,dx-\left\{\int\limits_0^1\sqrt{x}\,dx+\int\limits_1^4\sqrt{x}dx+......+\int\limits_{49}^{64}\sqrt{x}\,dx\right\}$

27. $I=\int\limits_0^{64}\sqrt[3]{x}\,dx-\left\{\int\limits_0^1\left[\sqrt[3]{x}\right]dx+\int\limits_1^8\left[\sqrt[3]{x}\right]dx+......+\int\limits_{27}^{64}\left[\sqrt[3]{x}\right]dx\right\}$

28. **Split integrals**

29 $I+I=\int\limits_{-3}^{3}x^8(1)dx \quad \therefore \{x\}-\{-x\}=1$

30. put $x=3\sin\theta$ and using reduction formula

31. Using reduction formula

32. $\sin^3x.\cos^2x$ is odd function and $\sin^2x.\cos^3x$ is an even function

33. Using formula $\int\limits_0^a f(x)dx=\int\limits_0^a f(a-x)dx$

34. $\because \sin^4 t+\cos^4+$ is periodic with period $\dfrac{\pi}{2}$

Now $f(x+\pi)=\int\limits_0^{\pi+x}\left(\sin^4 t+\cos^4 t\right)dt$

$=\int\limits_0^{x}\left(\sin^4 t+\cos^4 t\right)dt+\int\limits_x^{x+\pi}\left(\sin^4 t+\cos^4 t\right)dt$

$=f(x)+\int\limits_0^{\pi}\left(\cos^4 t+\sin^4 t\right)dt$

$=f(x)+2\int\limits_0^{\pi/2}\left(\cos^4 t+\sin^4 t\right)dt=f(x)+2f\left(\dfrac{\pi}{2}\right)$

35. $y=\sin x$, $\left(\dfrac{\pi}{2},1\right)$

36. $I_1-I_2=\int\limits_0^{\pi/2}(\cos\theta-\sin 2\theta)f\left(\sin\theta+\cos^2\theta\right)d\theta$

Put $\sin\theta+\cos^2\theta=t$

37. we have $f(x+y)=f(x)+f(y)$ for all $x,y\in R$

$\Rightarrow f(x)=\lambda$ x for all $x\in R$ where $\lambda=f(1)$

$\Rightarrow \int\limits_{-2}^{2}f(x)dx=0$

38. We have $I_0=\int\limits_0^1 e^x dx=e-1$ and for $n\ge 1$

$I_n=\left[e^x.(x-1)^n\right]_0^1-n\int\limits_0^1 e^x(x-1)^{n-1}dx$

$I_n=-(-1)^n-nI_{n-1}$

$\therefore I_1=1-(1)I_0=1-(e-1)=2-e$

$I_2=-1-2I_1=-1-2(2-e)=2e-5$

$I_3=1-3I_2=16-6e$

39. $e<x<4<e^2\Rightarrow x<e^2 \Rightarrow \log x<2$

40. Use by parts

41. put x = -t in Ist integration

42. $c_1\to c_1-(c_2+c_3) \quad \Rightarrow f(x)=\sin 3x$

43. $I=\int\limits_{\pi/4}^{\pi/2}2\sin t\cos t\,dt+\int\limits_{\pi/2}^{\pi}(-\sin t\cos t+\sin t\cos t)$

$dt+\int\limits_{\pi}^{5\pi/4}-2\sin t\cos t\;dt$

44. $0\le x\le 1\Rightarrow 1\le e^{x^2}\le e$

45. $0<x<1\Rightarrow 1-x^{2n}>1-x^2 \Rightarrow 1>\sqrt{1-x^{2n}}>\sqrt{1-x^2}$

46 $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{1}{\sqrt{2rn-r^2}}$

47. Given series $=e^{\underset{n\to\infty}{Lt}\frac{1}{n}\sum\limits_{r=1}^{n}\log(1+\frac{r}{n})}$

48. Given series $=e^{\underset{n\to\infty}{Lt}\frac{1}{n}\sum\limits_{r=1}^{n}\log(1+\frac{r^2}{n^2})}$